HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and optimized memory subsystem models for next-generation SoCs using **C++** and **SystemC**, focusing on coherency protocols and interconnect fabrics.
- Designed and implemented **Python** scripts for automating data processing and generating operational reports, reducing manual effort by approximately **40%**.
- Contributed to the architectural exploration of system-level designs to enhance performance and reduce power consumption in **SoC** development.
- Modeled memory infrastructure, including cache coherency protocols and **Network-on-Chip (NoC)** interconnects, to evaluate system-level performance.
- Wrote and maintained **shell scripts** for system administration tasks and log analysis on Linux environments, improving operational efficiency.
- Collaborated with cross-functional engineering teams to integrate client IP models into the overall **SoC architecture**, ensuring seamless development.
- Containerized system models using **Docker** for consistent deployment across development and testing environments.

SKILLS

Programming Languages – C++, Python, Shell Scripting

Modelling & Simulation – SystemC, Performance Modelling, System-Level Modelling

Computer Architecture – SoC Design, Memory Subsystem, Cache Architecture, CPU Architecture, GPU Architecture, Multimedia

Hardware Protocols – AXI, CHI, On-Chip Interconnect, Cache Coherency, Network-on-Chip (NoC)

Development Workflow – RTL Design, Hardware Development, CI/CD, Git, Code Reviews

Cloud & Containerization – Docker, AWS (EC2, S3, Lambda, CloudWatch)

PROJECTS

SoC Memory Controller Modelling (HCL Technologies)

- Developed a **SystemC** model for a memory controller architecture, simulating read/write operations and latency characteristics.
- Integrated the memory controller model with **AXI** interconnect fabric simulations to analyze data flow and identify potential bottlenecks.
- Analyzed simulation results to propose optimizations for **cache coherency** protocols, aiming to reduce system overhead by an estimated **15%**.

Python-based Performance Analysis Framework (HCL Technologies)

- Engineered a **Python** framework to parse and analyze simulation logs from complex hardware designs, extracting key performance metrics.
- Implemented data visualization tools within the framework to present performance trends and identify areas for **power optimization**.
- Automated the generation of summary reports, reducing analysis time for engineers by approximately **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)